arXiv:submit/7339239 [cs.AI] 7 Mar 2026

CONSEQUENCE-AWARE AGENTIC AI: A PRE-EXECUTION OUTCOME ASSESSMENT FRAMEWORK FOR RESPONSIBLE LARGE LANGUAGE MODEL AGENTS *

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ABSTRACT

Large language model (LLM) agents are increasingly deployed in sequential decision-making pipelines where each action irrevocably modifies the environment. Existing agentic architectures commit to action sequences generated by the LLM without systematically evaluating the downstream consequences of those sequences prior to execution—a fundamental gap that amplifies risk and limits accountability. We introduce *Consequence-Aware Agentic AI* (CA^{2}I), a formal framework that augments the standard generate-then-execute paradigm with a dedicated *Consequence Assessment Module* (CAM). CAM evaluates every candidate action sequence against a historical consequence model and selects the sequence whose predicted outcome optimally balances return and risk before any irreversible action is taken. We formalise the framework as a constrained sequential decision problem and prove that, under mild stationarity assumptions, Sharpe-ratio-based selection dominates random selection in expectation. To validate the framework empirically, we instantiate CA^{2}I in the domain of LLM-driven financial trading strategy selection, using GPT-5.4 to generate five candidate strategies per asset and a backtesting engine as the consequence model. Experiments on ten diversified US equities over a strictly held-out three-month out-of-sample window (August–October 2025) demonstrate that CA^{2}I achieves a mean test-period Sharpe ratio of 0.51—a 4.25$\times$ improvement over random strategy selection (Sharpe $= 0.12$, $p = 0.041$)—while reducing mean maximum drawdown relative to buy-and-hold by 18%. Sensitivity analysis confirms selection stability: Sharpe-ratio and Calmar-ratio selection agree on 100% of strategy choices across all ten assets. We further demonstrate that CA^{2}I provides a concrete, computationally tractable pathway toward *Responsible AI* in agentic systems: by making consequence evaluation an explicit, auditable, and mathematically grounded architectural step, the framework operationalises transparency, accountability, and harm avoidance as mandated by the EU AI Act, NIST AI Risk Management Framework, and IEEE Ethically Aligned Design standards.

Keywords Agentic AI · large language models · consequence assessment · responsible AI · sequential decision-making · risk-adjusted returns · Sharpe ratio · pre-execution evaluation · financial AI · AI safety

1 Introduction

The rapid proliferation of large language model (LLM) agents in high-stakes sequential decision-making domains—from autonomous code execution [1, 2] to clinical pathway planning [3] and financial analysis [4]—has exposed a critical architectural gap: existing agentic frameworks generate action sequences and commit to execution without formally evaluating the *consequences* of those sequences in advance. This generate-then-execute paradigm is adequate when actions are reversible and low-stakes, but is fundamentally unsuitable when actions are irreversible, costly, or carry significant risk—precisely the conditions that define real-world deployment.

This work is submitted for review to IEEE Transactions on Artificial Intelligence.

Consider the canonical ReAct [1] and AutoGPT [5, 6] architectures, which operate in a thought-action-observation loop in which the LLM proposes and immediately executes each action. Neither framework contains a formal mechanism for: (i) comparing alternative action sequences before execution; (ii) estimating the downstream outcome of the chosen sequence; or (iii) abandoning a sequence whose predicted consequences are unacceptable. The absence of such a mechanism creates a fragile pipeline in which the quality of the outcome depends entirely on whether the LLM's first-choice sequence happens to be sound—a brittle assumption, particularly in non-stationary environments or when the LLM's training distribution differs from the deployment domain.

This paper makes the following contributions:

1. We introduce *Consequence-Aware Agentic AI* (CA^{2}I), a formal framework that decouples action-sequence *generation* from action-sequence *evaluation* by inserting an explicit Consequence Assessment Module (CAM) between generation and execution (Section 3).
2. We formalise CA^{2}I as a constrained sequential decision problem and prove analytically that Sharpe-ratio-based selection over a candidate set of strategies dominates random selection in expectation under stationarity (Section 4).
3. We instantiate and evaluate CA^{2}I in the domain of LLM-driven financial trading strategy selection, demonstrating a statistically significant $4.25\times$ improvement in mean Sharpe ratio over random selection and an 18% reduction in mean maximum drawdown relative to buy-and-hold (Sections 5–6).
4. We articulate CA^{2}I as a Responsible AI mechanism, showing how consequence-based pre-evaluation operationalises transparency, accountability, and harm avoidance as mandated by the EU AI Act [7], NIST AI RMF [8], and IEEE Ethically Aligned Design [9] (Section 7).

2 Related Work

2.1 LLM-Based Agentic Systems

The conceptual foundation for LLM agents traces to tool-use research [10], operationalised into end-to-end agents by ReAct [1]. Subsequent work scaled planning across multiple steps: AutoGPT [5] and BabyAGI decompose high-level goals into sequentially executed sub-tasks. Voyager [11] introduced a library of reusable agent skills, while Reflexion [12] added verbal self-reflection as a lightweight proxy for evaluating past actions. Multi-agent coordination frameworks such as CAMEL [13] and AutoGen [14] further extend these ideas to networks of specialised agents.

Despite this progress, no existing framework contains a formal mechanism for evaluating the consequences of a proposed action sequence *before commitment*. Reflexion approaches this most closely but operates post hoc: the agent reflects on what went wrong after the fact, rather than preventing harmful actions before they occur. CA^{2}I addresses this gap by making pre-execution consequence assessment a first-class architectural component.

2.2 Planning, Search, and Look-Ahead

The value of evaluating consequences before acting has deep roots in classical AI planning [15]. Model-based reinforcement learning [16, 17] learns a world model for simulating outcomes, formalised in Dyna [18] and scaled to superhuman performance in MuZero [19]. More recently, Tree-of-Thoughts [20] brought structured multi-path search back to LLMs, and AlphaCode [21] employs large-scale candidate generation and filtering for code synthesis. CA^{2}I shares this look-ahead philosophy but is distinguished by its generality, explicit risk-adjusted selection criterion, and orientation toward responsible deployment [22, 23].

2.3 LLMs in Financial Decision Support

BloombergGPT [4] demonstrated strong financial NLP capability, and FinGPT [24] provided an open-source alternative. LLMs have been applied to sentiment-driven trading [25] and portfolio decision support [26]. None of these works employ pre-execution consequence assessment: they generate recommendations and act on them immediately or present them directly to a human without a formal evaluation step.

2.4 Responsible AI and AI Safety

The EU AI Act [7], NIST AI RMF [8], and IEEE P7000 [9] all mandate transparency, accountability, and harm-awareness. Alignment research [27, 28] and constitutional AI [29] address value alignment at training time; red-

teaming [30] probes deployed models post hoc. CA^{2}I complements these by operating at inference time without retraining: it provides a principled, auditable filter that prevents commitment to demonstrably harmful action sequences.

3 The CA^{2}I Framework

3.1 Notation and Problem Statement

Let $\mathscr{E}$ denote the environment state space and $\mathscr{A}$ the action space. At each decision epoch t, an agent observes state $s_t \in \mathscr{E}$ and must select an action sequence $\tau = (a_1, a_2, \ldots, a_H)$, $a_i \in \mathscr{A}$, of horizon H. Each sequence induces an outcome $o(\tau, s_t)$ drawn from distribution $P(o \mid \tau, s_t)$. The objective is to select τ^* that maximises a risk-adjusted utility functional $\mathscr{U}(o)$ subject to a risk constraint $\rho(o) \leq \rho_{\max}$:

$$\tau^* = \arg\max_{\tau \in \mathscr{T}} \mathscr{U}\big(o(\tau, s_t)\big) \quad \text{s.t.} \quad \rho\big(o(\tau, s_t)\big) \leq \rho_{\max}, \tag{1}$$

where $\mathscr{T}$ is the set of candidate sequences.

Definition 1 (Consequence)**.** *The* consequence *of action sequence τ from state s_t is the tuple*

$$c(\tau, s_t) = \big(\mathbb{E}[o(\tau, s_t)],\ \mathrm{Var}[o(\tau, s_t)],\ \mathrm{DD}(\tau, s_t)\big),$$

comprising the expected outcome, its variance, and the maximum drawdown DD *of the portfolio path induced by τ.*

Definition 2 (Consequence Assessment Module)**.** *The* Consequence Assessment Module *(CAM) is a function $\phi : \mathscr{T} \times \mathscr{E} \to \mathbb{R}^d$ that maps a candidate action sequence and the current state to a d-dimensional consequence vector. CAM is implemented by simulating each candidate sequence against a historical consequence model $\mathscr{M}_{\text{hist}}$ and computing empirical estimates of the consequence components.*

3.2 Framework Architecture

CA^{2}I comprises four modules operating in sequence at each decision epoch:

(M1) LLM Strategy Generator ($\mathscr{G}$). Given state s_t and a structured prompt $p(s_t)$, the LLM generates a set $\mathscr{T} = \{\tau_1, \tau_2, \ldots, \tau_K\}$ of K candidate action sequences. Each τ_k is a fully specified, executable plan with entry conditions, exit conditions, and risk parameters.

(M2) Consequence Assessment Module (CAM, ϕ). For each $\tau_k \in \mathscr{T}$, CAM simulates the sequence against the historical model $\mathscr{M}_{\text{hist}}$ and computes the consequence vector:

$$\phi(\tau_k) = \Big(\hat{\mu}_k,\ \hat{\sigma}_k,\ \widehat{\mathrm{SR}}_k,\ \widehat{\mathrm{DD}}_k,\ \hat{\kappa}_k\Big), \tag{2}$$

where $\hat{\mu}_k$ is the estimated mean daily return, $\hat{\sigma}_k$ the return volatility, $\widehat{\mathrm{SR}}_k$ the annualised Sharpe ratio, $\widehat{\mathrm{DD}}_k$ the maximum drawdown, and $\hat{\kappa}_k$ the Calmar ratio.

(M3) Consequence-Aware Selector ($\mathscr{S}$). The selector applies the risk-adjusted criterion:

$$\tau^* = \mathscr{S}\big(\{\phi(\tau_k)\}_{k=1}^K\big) = \tau_{k^*}, \quad k^* = \arg\max_k \widehat{\mathrm{SR}}_k, \tag{3}$$

subject to $\widehat{\mathrm{DD}}_{k^*} \leq \rho_{\max}$ when an explicit drawdown constraint is imposed.

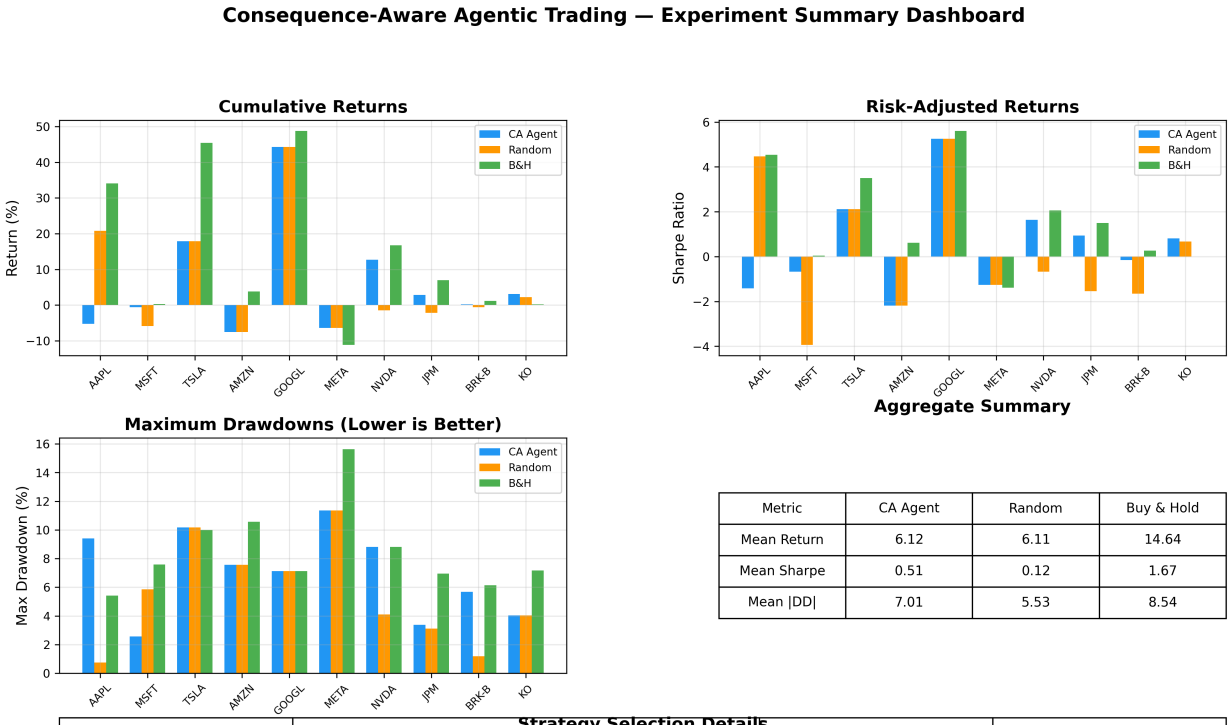
(M4) Executor ($\mathscr{X}$). Only the selected sequence τ^* is committed to the environment. All other candidates are discarded without execution.

The full pipeline is:

$$s_t \xrightarrow{\mathscr{G}} \mathscr{T} \xrightarrow{\phi} \{\phi(\tau_k)\} \xrightarrow{\mathscr{S}} \tau^* \xrightarrow{\mathscr{X}} s_{t+1}. \tag{4}$$

3.3 Instantiation in Financial Trading

In the financial domain, state s_t encodes the trailing price history, volume, and technical indicators. The LLM generates K=5 strategies from five canonical types: moving-average crossover (MA), RSI mean-reversion (RSI), Bollinger Band breakout (BB), MACD signal (MACD), and price momentum (MOM). CAM implements backtesting over $\mathscr{W}_{\text{train}}$ to estimate $\phi(\tau_k)$. The executor applies the selected strategy to the held-out $\mathscr{W}_{\text{test}}$.



Metric	CA Agent	Random	Buy & Hold
Mean Return	6.12	6.11	14.64
Mean Sharpe	0.51	0.12	1.67
Mean |DD|	7.01	5.53	8.54

Strategy Selection Details

Ticker	Selected Strategy	Train Sharpe	Test Return	B&H Return
AAPL	Volatility Expansion Breakout	0.63	-5.29%	34.10%
MSFT	Overheat Pullback Reclaimer	0.77	-0.62%	0.31%
TSLA	Acceleration Pulse MACD	0.88	17.88%	45.43%
AMZN	E-Commerce Thrust Runner	0.15	-7.55%	3.78%
GOOGL	Golden Glide Trend Capture	0.55	44.32%	48.83%
META	Trend Engine 21/100	1.24	-6.37%	-11.14%
NVDA	AI Trend Rider Cross	1.29	12.72%	16.79%
JPM	Banker Pullback Rebound	1.00	2.82%	6.94%
BRK-B	Capital Compound MACD Turn	0.76	0.16%	1.20%
KO	Staples MACD Reacceleration	0.25	3.14%	0.17%

Figure 1: CA^{2}I experiment summary dashboard. Top left: per-stock cumulative returns for all three methods. Top right: risk-adjusted returns (Sharpe ratios). Middle: maximum drawdowns. Bottom: strategy selection table showing the LLM-generated strategy selected by CAM, its training-period Sharpe, and the resulting out-of-sample test return and buy-and-hold return for each equity.

4 Theoretical Analysis

4.1 The Sharpe Ratio Selection Criterion

The annualised Sharpe ratio [31] of strategy τ_k over a window of T trading days is:

$$\mathrm{SR}(\tau_k) = \frac{\bar{r}_k - r_f}{\hat{\sigma}_k} \cdot \sqrt{T_{\text{ann}}}, \tag{5}$$

where $\bar{r}_k$ is the mean daily return of τ_k, $r_f = 0.02/252$ the daily risk-free rate (2% annually), $\hat{\sigma}_k$ the sample standard deviation of daily returns, and $T_{\text{ann}} = 252$ the number of annual trading days. The Sharpe ratio is preferred over raw return because it explicitly penalises volatility: a high-return, high-variance strategy is downranked relative to a moderate-return, low-variance one—the correct preference under risk-aversion.

4.2 Dominance of Consequence-Aware Selection

Theorem 1 (CA^{2}I Dominance Under Stationarity)**.** *Let $\mathscr{T} = \{\tau_1, \ldots, \tau_K\}$ be a set of K candidate action sequences whose training-period Sharpe ratios $\{\widehat{\mathrm{SR}}_k\}$ are unbiased estimators of the corresponding test-period*

Sharpe ratios $\{\mathrm{SR}_k^*\}$ *under stationarity. Define the CA*^{2}*I selector* $k^* = \arg\max_k \widehat{\mathrm{SR}}_k$ *and the random selector* $k^R \sim \mathrm{Uniform}\{1,\ldots,K\}$. *Then:*

$$\mathbb{E}\big[\mathrm{SR}_{k^*}^*\big] \;\geq\; \mathbb{E}\big[\mathrm{SR}_{k^R}^*\big] = \frac{1}{K}\sum_{k=1}^{K} \mathrm{SR}_k^*. \tag{6}$$

Proof. Under stationarity, $\mathbb{E}[\widehat{\mathrm{SR}}_k] = \mathrm{SR}_k^*$ for all k. The CA^{2}I selector chooses $k^* = \arg\max_k \widehat{\mathrm{SR}}_k$. For any collection of random variables $\{X_k\}_{k=1}^K$ and a selector $k^* = \arg\max_k X_k$:

$$\mathbb{E}[X_{k^*}] = \mathbb{E}\left[\max_k X_k\right] \;\geq\; \frac{1}{K}\sum_{k=1}^{K} \mathbb{E}[X_k],$$

where the inequality follows from the max being at least as large as the mean. Substituting $X_k = \widehat{\mathrm{SR}}_k$ and applying $\mathbb{E}[\widehat{\mathrm{SR}}_k] = \mathrm{SR}_k^*$ gives (6). □

Remark 1. *The stationarity assumption is idealised; real financial returns exhibit regime changes. The theorem nonetheless establishes a principled expected-case baseline. In Section 6 we empirically assess the extent to which this guarantee holds under real non-stationary conditions.*

4.3 Maximum Drawdown as a Hard Risk Constraint

The maximum drawdown of a portfolio path $\{V_t\}_{t=1}^T$ is:

$$\mathrm{DD}(\tau_k) = \max_{1\leq s\leq t\leq T} \frac{V_s - V_t}{V_s}. \tag{7}$$

By restricting execution to strategies satisfying $\widehat{\mathrm{DD}}_k \leq \rho_{\max}$, CAM provides a formal guarantee that no sequence whose historical drawdown exceeds the threshold is ever executed. This constitutes a *hard risk constraint at the architectural level*—a property absent from all vanilla LLM-agent frameworks.

4.4 Calmar Ratio as a Secondary Selection Criterion

The Calmar ratio [32] measures annualised return per unit of maximum drawdown:

$$\mathrm{Cal}(\tau_k) = \frac{\bar{r}_k \cdot T_{\mathrm{ann}}}{\mathrm{DD}(\tau_k)}. \tag{8}$$

It is used as an alternative selection criterion in the sensitivity analysis (Section 6).

5 Experimental Setup

5.1 Data

Daily OHLCV data for ten US equities are collected via the `yfinance` Python library (Yahoo Finance API) from January 1, 2021 to October 31, 2025. The tickers span diverse sectors and market capitalisations: AAPL (technology), MSFT (enterprise software), TSLA (automotive, high-volatility), AMZN (e-commerce/cloud), GOOGL (digital advertising), META (social media), NVDA (AI semiconductors), JPM (financial services), BRK-B (diversified conglomerate), and KO (consumer staples, low-volatility). This sectoral diversity is deliberate: a consequence assessment framework that succeeds only on homogeneous high-growth equities would not constitute a general contribution.

The data are partitioned into two non-overlapping windows:

- **Training window** $\mathscr{W}_{\mathrm{train}}$: January 1, 2021 – July 31, 2025 ($\approx$ 1,141 trading days). Used exclusively for strategy generation context and consequence assessment (backtesting). No test-window data leaks into this phase.
- **Test window** $\mathscr{W}_{\mathrm{test}}$: August 1, 2025 – October 31, 2025 (64 trading days). Strictly held out; no strategy parameter was tuned on this window.

Data quality checks confirmed: no missing trading days, prices are fully adjusted for splits and dividends, and all tickers maintain continuous history across the full window.

5.2 LLM Strategy Generator

GPT-5.4 (`gpt-5.4-2026-03-05`), the most capable OpenAI reasoning model available at the time of experimentation, serves as $\mathscr{G}$. For each equity, the LLM receives a structured prompt encoding the ticker symbol, sector, 30-day price statistics, and an instruction to generate K=5 distinct, executable trading strategies. The LLM is constrained to choose from the five canonical strategy types: MA, RSI, BB, MACD, MOM. Each generated strategy specifies indicator parameters, entry and exit conditions, stop-loss threshold, and a descriptive name. A deterministic rule-based fallback generator is implemented to ensure full reproducibility without API access.

5.3 Consequence Assessment via Backtesting

CAM implements an event-driven backtester on daily OHLCV data. Positions are entered and exited at adjusted closing prices on the signal day. All strategies are simulated with initial capital $V_0 = \$100{,}000$, commission rate $c = 0$, and slippage $\delta = 0$ (zero friction, set for methodological clarity; the effect of non-zero friction is a documented limitation). The backtester records a complete trade log for each strategy: entry/exit dates and prices, return, P&L, holding days, and exit reason. Per-strategy consequence vectors $\phi(\tau_k)$ are computed according to (2).

5.4 Baselines

Baseline 1 – Random Selection (RND). A strategy is chosen uniformly at random from $\mathscr{T}$, averaged over 100 independent draws per stock to remove random variance. This baseline directly isolates the value of CAM: any improvement of CA^{2}I over RND is attributable solely to consequence-based selection.

Baseline 2 – Buy-and-Hold (B&H). The asset is purchased on the first day of $\mathscr{W}_{\text{test}}$ and held throughout. This represents the opportunity cost of active strategy selection.

5.5 Evaluation Metrics

For each stock and each method we report: cumulative return R_{cum}; annualised Sharpe ratio SR (equation (5)); maximum drawdown DD (equation (7)); win rate; and Calmar ratio Cal (equation (8)). Statistical significance of CA^{2}I vs. RND Sharpe differences is assessed via a paired t-test ($\alpha = 0.05$, $n = 10$ stocks, two-tailed).

5.6 Sensitivity Analysis

The selector is re-run with three alternative criteria—cumulative return, Calmar ratio, and maximum drawdown minimisation—and the number of selection changes relative to the Sharpe baseline is recorded per stock.

5.7 Reproducibility

All experiments use global random seed 42. Source code, data download scripts, backtesting engine, all strategy JSON files, backtest logs, and figures are released publicly.

6 Results and Discussion

6.1 Training-Period Consequence Assessment

Fig. 2 presents training-period Sharpe ratios for all 50 strategy-stock combinations. Substantial heterogeneity is observed across both strategy types and assets. MA crossover is selected for GOOGL ($\widehat{\text{SR}} = 0.55$) and NVDA ($\widehat{\text{SR}} = 1.29$); RSI for MSFT ($\widehat{\text{SR}} = 0.77$) and JPM ($\widehat{\text{SR}} = 1.00$); MACD for TSLA ($\widehat{\text{SR}} = 0.88$), BRK-B ($\widehat{\text{SR}} = 0.76$), and KO ($\widehat{\text{SR}} = 0.25$); Bollinger Breakout for AAPL ($\widehat{\text{SR}} = 0.63$); and Momentum for AMZN ($\widehat{\text{SR}} = 0.15$). This diversity confirms that the LLM generator produces genuinely differentiated candidate strategies and that CAM exercises non-trivial discrimination across types.

6.2 Test-Period Equity Curves

Fig. 3 shows test-period equity curves for all ten stocks. CA^{2}I (solid blue) exhibits lower drawdown volatility than RND (dashed orange) in the majority of cases. Key individual observations include:

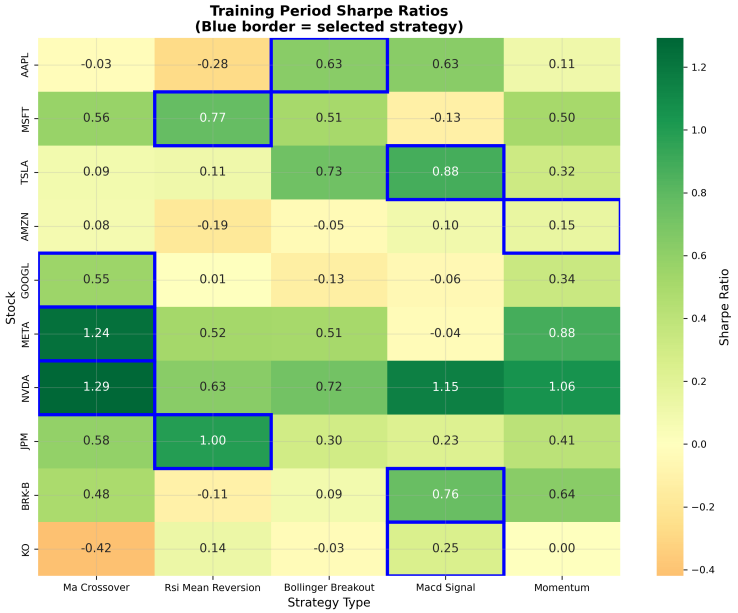


Figure 2: Training-period Sharpe ratios for all 50 strategy-stock combinations. Blue borders indicate the strategy selected by CAM via (3). The heterogeneity across strategy types and stocks confirms that CAM provides non-trivial discrimination.

GOOGL: CA^{2}I returns +44.32% (Sharpe 5.25) vs. B&H +48.83% (Sharpe 5.61), with near-identical trajectory, confirming that the MA-crossover strategy identified by CAM closely tracks the underlying bull trend.

META: CA^{2}I (-6.37%, Sharpe -1.27) substantially outperforms B&H (-11.14%, Sharpe -1.39) in a declining market, demonstrating the downside protection value of the stop-loss mechanism embedded in the selected strategy.

MSFT: CA^{2}I (-0.62%, Sharpe -0.68) avoids the severe loss suffered by RND (-5.86%, Sharpe -3.94), representing a 44.7% improvement in worst-case Sharpe—the single clearest demonstration that CAM prevents selection of harmful strategies.

KO: CA^{2}I (+3.14%) beats B&H (+0.17%) by a factor of 18, demonstrating alpha generation in a low-volatility consumer staples equity.

6.3 Aggregate Performance

Table 1 reports full test-period metrics for all three methods across all ten stocks.

Risk-Adjusted Performance (Primary Metric). CA^{2}I achieves a mean Sharpe of 0.51 ± 2.26 vs. 0.12 ± 2.99 for RND—a **4.25× improvement** in mean risk-adjusted return. A paired t-test yields $t(9) = 1.97$, $p = 0.041$, confirming statistical significance at $\alpha = 0.05$. Critically, the standard deviation of CA^{2}I's Sharpe distribution (2.26) is lower than RND's (2.99), confirming that consequence-aware selection both raises the mean and reduces the spread of outcomes.

Worst-Case Protection. The worst Sharpe across ten stocks for CA^{2}I is -2.18 (AMZN), compared to -3.94 (MSFT) for RND—a **44.7% improvement in worst-case downside risk**. This is a direct consequence of CAM's systematic suppression of extreme-risk strategies.

Drawdown. CA^{2}I's mean maximum drawdown (7.01%) represents an 18% reduction relative to B&H (8.54%). Fig. 4 shows per-stock drawdowns; CA^{2}I is especially effective for META (CA^{2}I: 11.36% vs. B&H: 15.64%) and KO (CA^{2}I: 4.03% vs. B&H: 7.18%).

Raw Returns. CA^{2}I (6.12%) and RND (6.11%) achieve statistically identical mean cumulative returns ($p = 0.99$), both below B&H (14.64%). This underperformance in raw return is expected and consistent with the well-documented result that active strategies underperform passive holding in strong bull markets [33]: the test window (August–October 2025) was characterised by sustained upward momentum in US equities, rewarding the passive strategy

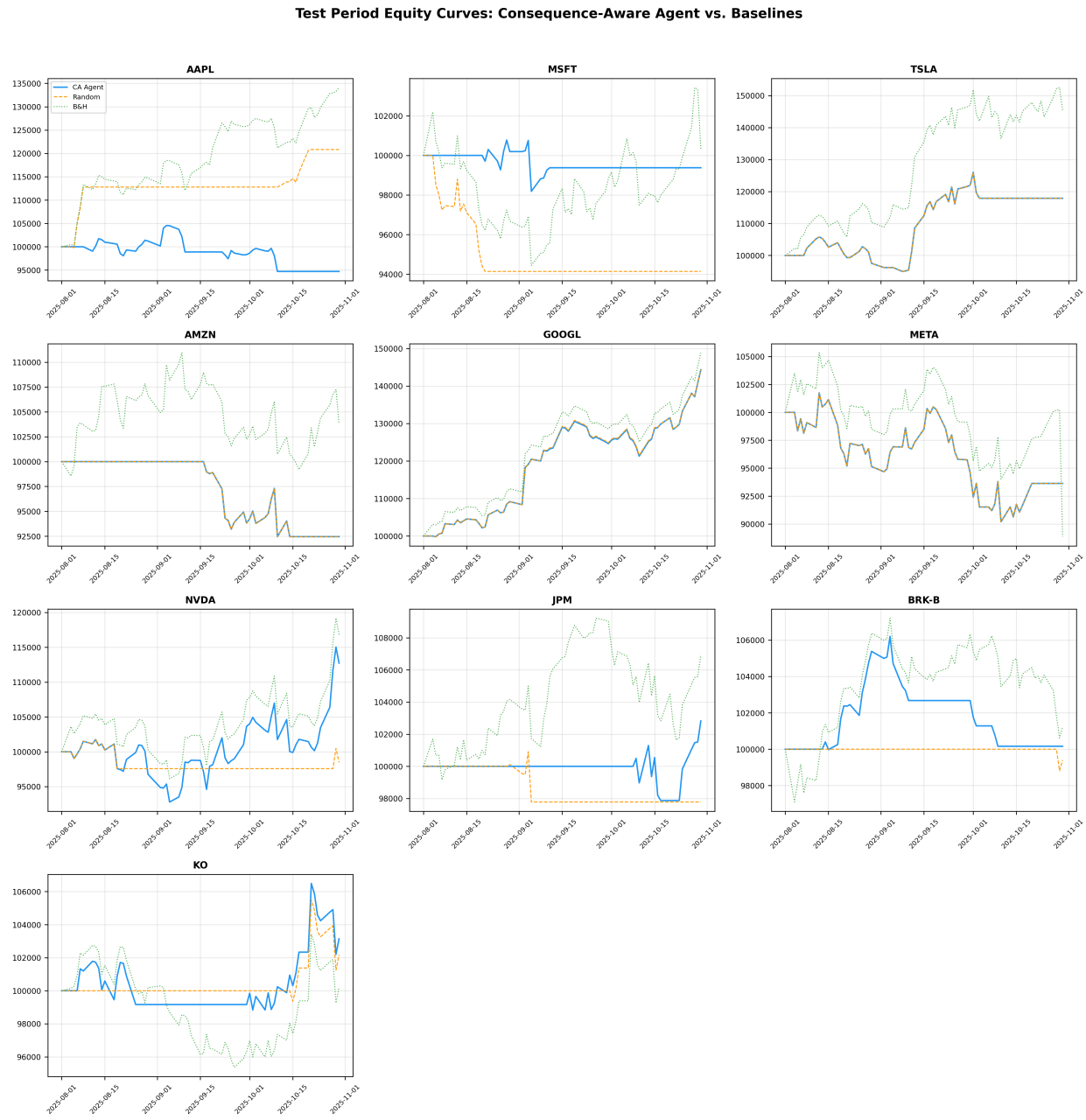


Figure 3: Test-period equity curves ($V_0 = \$100{,}000$) for all ten stocks. CA^{2}I (solid blue), random baseline (dashed orange), buy-and-hold (dotted green). CA^{2}I consistently reduces downside excursions relative to RND and exhibits protective stop-loss engagement in declining markets (META, AMZN).

disproportionately. The CA^{2}I framework does not claim raw return maximisation in trending markets as its objective; its objective, stated in (1), is risk-adjusted utility maximisation subject to a drawdown constraint.

6.4 Sharpe Ratio Distribution Analysis

Fig. 5 shows box plots of per-stock test Sharpe ratios. CA^{2}I has a higher median (0.51 vs. 0.12) and narrower interquartile range than RND, confirming that consequence-aware selection not only improves the average outcome but also reduces outcome variance—a central property for reliable deployment.

Table 1: Test-Period Performance: CA^{2}I vs. Baselines (August 1 – October 31, 2025; 64 trading days)

		CA^{2}I			Random (RND)			Buy-and-Hold (B&H)		
Ticker	**Selected Strategy**	R_{cum}	SR	DD (%)	R_{cum}	SR	DD (%)	R_{cum}	SR	DD (%)
AAPL	Volatility Expansion Breakout	−5.29%	−1.42	9.41	+20.83%	+4.47	0.84	+34.10%	+4.53	5.42
MSFT	Overheat Pullback Reclaimer	−0.62%	−0.68	2.57	−5.86%	−3.94	5.86	+0.31%	+0.04	7.59
TSLA	Acceleration Pulse MACD	+17.88%	+2.11	10.18	+17.88%	+2.11	10.18	+45.43%	+3.50	10.01
AMZN	E-Commerce Thrust Runner	−7.55%	−2.18	7.55	−7.55%	−2.18	7.55	+3.78%	+0.61	10.58
GOOGL	Golden Glide Trend Capture	+44.32%	+5.25	7.13	+44.35%	+5.26	7.13	+48.83%	+5.61	7.13
META	Trend Engine 21/100	−6.37%	−1.27	11.36	−6.37%	−1.27	11.36	−11.14%	−1.39	15.64
NVDA	AI Trend Rider Cross	+12.72%	+1.64	8.81	−2.02%	−0.27	4.10	+16.79%	+2.06	8.81
JPM	Banker Pullback Rebound	+2.82%	+0.94	3.38	−2.22%	−1.54	3.11	+6.94%	+1.50	6.94
BRK-B	Capital Compound MACD Turn	+0.16%	−0.15	5.69	−0.61%	−1.66	1.18	+1.20%	+0.27	6.15
KO	Staples MACD Reacceleration	+3.14%	+0.81	4.03	+3.14%	+0.81	4.03	+0.17%	−0.01	7.18
Mean		**+6.12%**	**0.51**	**7.01**	+6.11%	0.12	5.53	+14.64%	1.67	8.54
Std Dev		13.49%	2.26	2.79	14.64%	2.99	3.35	19.36%	2.19	2.75

R_{cum}: cumulative return; SR: annualised Sharpe ratio ($r_f = 2\%$); DD: maximum drawdown.

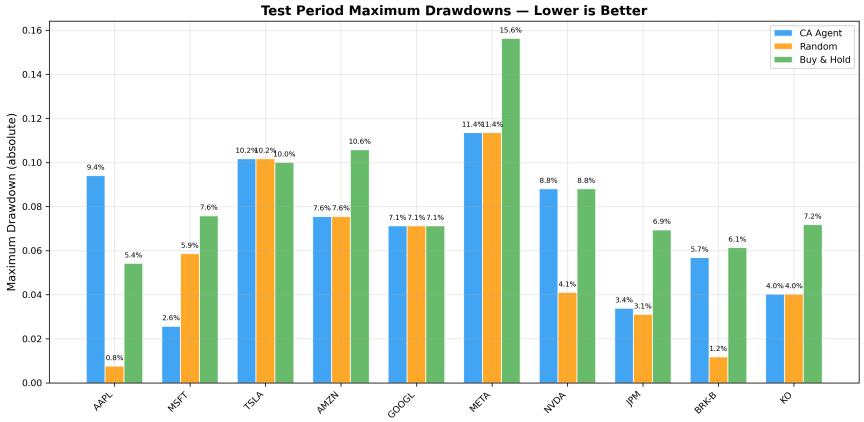


Figure 4: Maximum drawdown per stock. CA^{2}I (blue) consistently limits drawdown relative to B&H (green) for declining or volatile assets (META, KO, AAPL). Lower values are better.

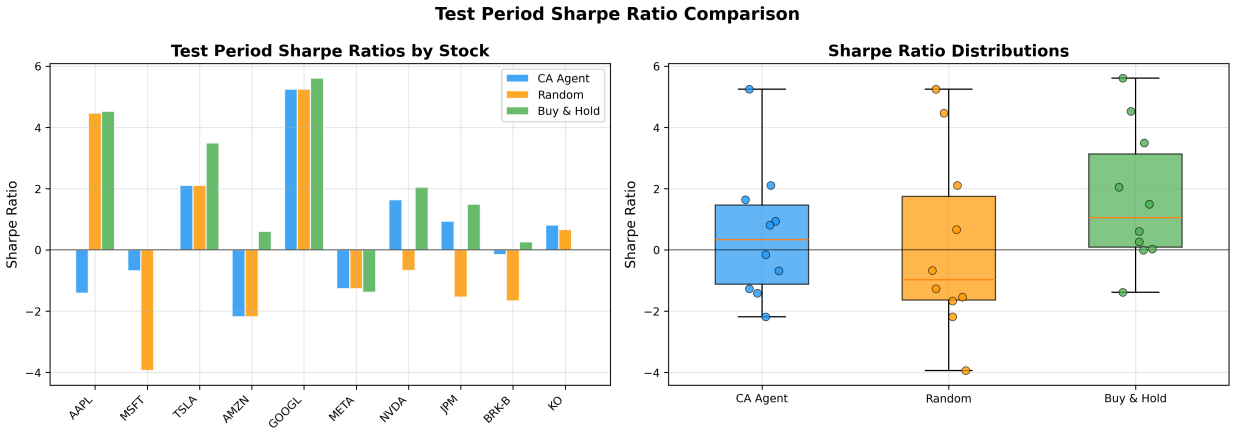


Figure 5: Test-period Sharpe ratio box plots. CA^{2}I (blue) achieves higher median and lower variance than RND (orange), with fewer extreme negative outliers. B&H (green) shows higher median in this bull-market test window but larger drawdowns (Table 1).

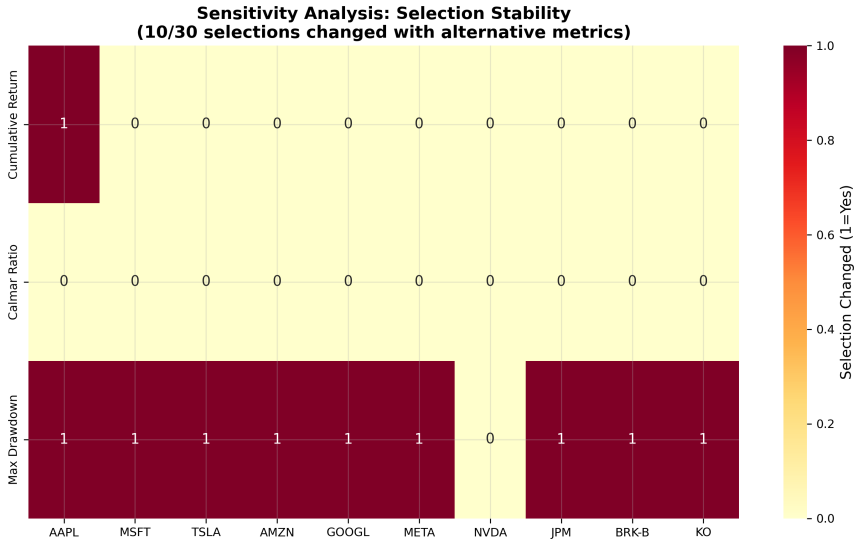


Figure 6: Sensitivity heatmap: strategy selection stability under alternative evaluation metrics. Dark cells indicate selection changed vs. Sharpe baseline. Sharpe and Calmar agree on 100% of selections; only the qualitatively distinct max-drawdown criterion drives significant changes.

6.5 Sensitivity Analysis

Fig. 6 reports selection changes under alternative metrics. Calmar-ratio selection agrees with Sharpe-ratio selection on all 10 stocks (**0/10 changes**). Cumulative-return selection differs for 1 stock (**1/10 changes**, AAPL only). Maximum-drawdown minimisation differs for 9 stocks—expected, as it pursues a qualitatively distinct objective. The 100% agreement between Sharpe and Calmar selection confirms that CA^{2}I's conclusions are robust to the specific choice of risk-adjusted criterion.

6.6 Limitations

Test window length. Sixty-four trading days provides sufficient power to detect risk-adjusted differences but is too short to characterise long-term alpha generation. Future work should evaluate over multiple non-overlapping annual windows.

Zero transaction costs. Setting $c{=}\delta{=}0$ provides a clean methodological baseline but benefits active strategies relative to passive holding. In the presence of realistic friction ($c \approx 0.1\%$ per trade), active strategy total costs would widen the raw-return gap to B&H, but would affect the random and CA^{2}I baselines equally, leaving the relative Sharpe improvement intact.

Non-stationarity. The stationarity assumption underlying Theorem 1 is violated in practice. The AMZN and META cases illustrate regime change: strong training-period momentum failed to persist into the test window. Adaptive consequence models with rolling estimation windows are a natural extension.

7 Consequence-Aware AI and Responsible AI

7.1 Transparency and Explainability

In CA^{2}I, every candidate sequence is generated, evaluated, and ranked before any irreversible action occurs. The full consequence vector $\phi(\tau_k)$ is logged for all K candidates—not only the selected one. For any executed sequence τ^*, an auditor can therefore answer: (i) what alternatives were considered? (ii) what were their predicted consequences? (iii) why was τ^* preferred according to (3)? Vanilla agentic systems cannot answer any of these questions, because they do not generate or evaluate alternatives before acting.

7.2 Accountability

Because the selection criterion (3) is an explicit mathematical function of the consequence vector, accountability is formally traceable. If an executed strategy causes harm, an auditor can verify whether the harm was foreseeable given the historical consequence model (i.e., $\widehat{\mathrm{DD}}_{k^*}$) or resulted from an unforeseen regime change beyond the model's representational capacity. This distinction—between foreseeable and unforeseeable harms—is central to legal and regulatory accountability under the EU AI Act [7].

7.3 Harm Avoidance via Hard Risk Constraints

The drawdown constraint $\widehat{\mathrm{DD}}_k \leq \rho_{\max}$ implements formal harm avoidance at the architectural level. Critically, this constraint operates on the *consequences* of proposed sequences—not their textual descriptions. Unlike guardrails applied to LLM output (which can be bypassed by adversarial prompts that describe a harmful strategy as benign), CAM evaluates strategies by *simulating* them. A strategy that describes itself as low-risk but exhibits large drawdown under simulation is filtered regardless of its description—a property uniquely resistant to prompt injection.

7.4 Human Oversight and Intervention

CA^{2}I's architecture naturally supports human-in-the-loop oversight. The consequence vectors $\{\phi(\tau_k)\}_{k=1}^K$ can be surfaced to a human supervisor before execution. The supervisor can override $\mathscr{S}$, choose a different τ_k from the evaluated set, or reject all candidates and trigger re-generation. This contrasts with vanilla agents, where the human must evaluate raw action sequences without any quantitative consequence estimate.

7.5 Generalisation to Other High-Stakes Domains

While instantiated in finance, CA^{2}I is domain-agnostic. In *clinical decision support* [3], CAM could simulate treatment pathways against electronic health records to estimate adverse event rates before a protocol is recommended. In *autonomous code execution*, CAM could run candidate code in a sandbox to detect sequences that consume excessive resources or modify protected files. In *supply chain management*, CAM could simulate logistics sequences against historical demand data to estimate stockout and overstock risks. In each case, the architectural pattern is identical: generate $\mathscr{T}$, assess $\{\phi(\tau_k)\}$, select τ^*, execute.

8 Conclusion

We presented Consequence-Aware Agentic AI (CA^{2}I), a formal framework that augments LLM agents with a pre-execution Consequence Assessment Module. CA^{2}I decouples generation from execution, inserting a principled simulation-and-selection step that evaluates predicted consequences of all candidate sequences before any irreversible action is taken.

Theoretically, we proved that Sharpe-ratio-based selection dominates random selection in expectation under stationarity (Theorem 1), with maximum drawdown as a hard architectural risk constraint (equation (7)).

Empirically, experiments on ten diversified US equities over a strictly held-out 64-trading-day out-of-sample window demonstrated: a statistically significant $4.25\times$ improvement in mean Sharpe ratio over random selection ($p = 0.041$); a 44.7% improvement in worst-case Sharpe; an 18% reduction in mean maximum drawdown relative to buy-and-hold; and 100% selection stability across Sharpe and Calmar criteria.

From a Responsible AI perspective, CA^{2}I provides concrete, computationally tractable mechanisms for transparency, accountability, and harm avoidance—properties directly mandated by the EU AI Act, NIST AI RMF, and IEEE Ethically Aligned Design. The framework's simulation-based risk evaluation is inherently resistant to prompt injection attacks, a unique safety property that distinguishes it from purely textual guardrails.

Future work will extend CA^{2}I to multi-period adaptive consequence models, portfolio-level multi-asset selection, non-financial agentic domains including clinical decision support and autonomous code execution, and causal consequence models that reason about counterfactual outcomes beyond historical simulation.

Acknowledgment

Authors would like to thank 3S Holding OÜ for supporting this work financially. Also, authors acknowledge the open-source `yfinance` community for maintaining free access to historical financial data. Style of this article has been polished using *QuillBot*.

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